# Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to: de Bono J, Mateo J, Fizazi K, et al. Olaparib for metastatic castration-resistant prostate cancer. N Engl J Med 2020;382:2091-102. DOI: 10.1056/NEJMoa1911440

# Supplementary Appendix

## Olaparib for Metastatic Castration-resistant Prostate Cancer

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# Contents

## PROfound investigators

The table below lists the principal investigator for each site who participated in the study and the number of patients enrolled.

| Country | Principal Investigator | Enrollment (N=387) |
|---|---|---|
| Argentina | Gabriela Gatica, Diego Kaen, Ernesto Korbenfeld, Luis Montes de Oca, Sandra Anabel Ostoich, Juan Sade | 9 |
| Australia | Pretoria Bilinski, Craig Gedye, Howard Gurney, Elizabeth Hovey, Edmond Kwan, Siobhan Ng, Shahneen Sandhu, Lavinia Spain, Aneta Suder, Thean Hsiang Tan | 24 |
| Austria | Richard Greil, Gero Kramer, Wolfgang Loidl, Johannes Meran, Herbert Stöger | 6 |
| Brazil | Felipe Cruz, Diogo da Rosa, Charles de Padua, Gabriel dos Anjos, Adilson Faccio, Andre Fay, Marcelo Freitas, Gustavo Girotto, Fabrício Augusto Oliveira, Suelen Martins, Marcelo Salgado, Fabio Schutz, Rodrigo Villarroel, Luis Zucca | 14 |
| Canada | Richard Casey, Kim Chi, Susan Ellard, Urban Emmenegger, Antonio Finelli, Sebastien Hotte, Nayyer Iqbal, Michael Kolinsky, Louis Lacombe, Jean-Benoit Paradis, Fred Saad, Simon Tanguay | 32 |
| Denmark | Steinbjørn Hansen | 1 |
| France | Jerome Alexandre, Emmanuelle Bompas, Karim Fizazi, Aude Flechon, Gwenaelle Gravis, Florence Joly, Vincent Massard, Loïc Mourey, Nicolas Penel, Guilhem Roubaud, Antoine Thiery-Vuillemin, Diego Tosi, Jean-Marc Tourani | 62 |
| Germany | Manfred Binder, Thomas Ebert, Susan Feyerabend, Jochen Gleissner, Marc-Oliver Grimm, Carsten Grüllich, Eva Hellmis, Wolfgang Hölzer, Jörg Klier, Stefan Machtens, Gerald Mickisch, Günter Niegisch, Martin Schostak, Arnulf Stenzl, Henrik Suttmann | 12 |
| Israel | Raanan Berger, Stephen Frank, Daniel Kejzman, Avivit Peer, Eli Rosenbaum, Avishay Sella | 9 |
| Italy | Rossana Berardi, Orazio Caffo, Ugo de Giorgi, Gaetano Facchini, Alketa Hamzaj, Emanuele Naglieri, Franco Nolè, Giuseppe Procopio, Roberto Sabbatini, Francesca Valcamonico | 15 |
| Japan | Shin Egawa, Masatoshi Eto, Yasuhisa Fujii, Kiyohide Fujimoto, Takayuki Goto, Katsuyoshi Hashine, Masashi Kato, Satoru Kawakami, Yukihiro Kondo, Nobuaki Matsubara, Ryuji Matsumoto, Masato Matsuzaki, Atsushi Mizokami, Masayoshi Nagata, Kazuo Nishimura, Chikara Ohyama, Takatsugu Okegawa, Mototsugu Oya, Hideki Sakai, Norihito Soga, Mikio Sugimoto, Hiroyoshi Suzuki, Kazuhiro Suzuki, Kenichi Tabata, Satoshi Tamada, Naoki Terada, Hiroji Uemura, Motohide Uemura, Junji Yonese, Kazuhiro Yoshimura | 57 |
| Norway | Daniel Heinrich | 3 |
| South Korea | Young Deuk Choi, Byung Ha Chung, Hong Koo Ha, Jun Hyuk Hong, Seong Soo Jeon, Jae Young Joung, Taek Won Kang, Tae Gyun Kwon, Ji Youl Lee | 29 |
| Spain | Juan Antonio Virizuela Echaburu, Carlos Álvarez Fernández, Joan Carles Galcerán, Daniel Castellano Gauna, Nuria Sala González, David Olmos Hidalgo, Javier Puente Vázquez | 9 |
| Sweden | Enrique Castellanos, Ingela Lissbrant | 2 |
| Taiwan | Chung-Hsin Chen, Po-Hui Chiang, Hsiao-Jen Chung, See-Tong Pang, Wen-Pin Su, Pai-Fu Wang, Shian-Shiang Wang, Wen-Jeng Wu, Su-Peng Yeh | 15 |

| | | |
|---|---|---|
| The Netherlands | Andries Bergman, Niven Mehra, Susanne Osanto, Metin Tascilar, Robbert van Alphen, Hendrik van den Berg, | 21 |
| Turkey | Cagatay Arslan, Irfan Cicin, Mustafa Erman, Fatih Kose, Nil Mandel, Mustafa Ozguroglu, Ali Sendur, Yuksel Urun | 30 |
| United Kingdom | Johann de Bono, Paul Elliott, Stephanie Gibbs, Mark Linch, Jonathan Shamash | 4 |
| United States | Neeraj Agarwal, Andrew Armstrong, Vasileios Assikis, James Bailen, Michael Carducci, William Clark, Sarah Colonna, Gregory Daniels, Paromita Datta, Nancy Davis, Nancy Dawson, Tanya Dorff, Curtis Dunshee, Michael Goodman, Theodore Gourdin, Ivan Grunberger, Maha Hussain, Benjamin Martin, Bryan Mehlhaff, Gregory Merrick, Gregg Newman, Luke Nordquist, Yeun-Hee Park, Kalpesh Patel, Christopher Pieczonka, Daniel Saltzstein, Oliver Sartor, Stephen Savage, Peter Schlegel, Ian Schnadig, David Shaffer, Neal Shore, Kelly Stratton, Ronald Tutrone, Daniel Vaena, Nicholas Vogelzang, Jingsong Zhang | 33 |

# Methods

## Biomarker algorithm for companion diagnostic development

The 15 pre-specified genes in the investigational clinical trial patient selection assay are a subset of the genes that can be tested using the FoundationOne® CDx (F1CDx) device. F1CDx is a next-generation-sequencing-based in vitro diagnostic device for detection of substitutions, insertion and deletion alterations (indels) and copy number alterations in 324 genes and select gene rearrangements, as well as genomic signatures including microsatellite instability and tumor mutational burden using DNA isolated from formalin-fixed paraffin-embedded tumor tissue specimens. The 15 qualifying genes for this study were selected based on mechanistic role in homologous recombination repair (HRR), clinical efficacy data, involvement in hereditary cancer risk, preclinical evidence of sensitivity to poly(ADP-ribose) polymerase (PARP) inhibition, prevalence across solid tumor types, and genetic reversion events that restore gene function in tumors from patients who have developed clinical resistance to PARP inhibitors.[1-5]

For each tumor specimen that passed tissue sample and sequencing quality controls, a clinical trial assay report was generated specifying the presence or absence of qualifying gene alterations. A patient had a qualifying alteration if any deleterious or suspected deleterious alteration was found in the 15 pre-specified genes with a direct or indirect role in HRR. An alteration was regarded as deleterious if it results in protein truncation (which includes nonsense, frameshift, or consensus splice site alterations), or select missense alterations well-known to be deleterious in ClinVar/BIC databases. Furthermore, larger-scale alterations, such as genomic truncating rearrangements or homozygous deletions, were also classified as qualifying. Patients without qualifying gene alterations as determined by the investigational patient selection assay were not eligible for the study.

## Sample requirements for prospective diagnostic testing

All samples were tested centrally at Foundation Medicine Inc. (Cambridge, MA, USA). Clinical study sites were requested to submit tumor tissue that was formalin fixed and paraffin embedded in the format of either a block or approximately 20 x 4 to 5-µm tissue sections. An accompanying original hematoxylin and eosin (H&E) stained tissue section was requested at sample submission, but if not available, H&E staining was performed upon receipt of the sample. A minimum tissue surface area of 5–7.5 mm$^2$ and a minimum of 20% tumor content was required for testing. Upon review of the H&E-stained tissue, a pathologist determined the amount of tissue that was required for DNA extraction based on a tumor tissue volume calculation.

### Study assessments

Time to pain progression was based on Brief Pain Inventory-Short Form item 3 ('worst pain in 24 hours) and initiation of opiate analgesic use (Analgesic Quantification Algorithm score). Time to first symptomatic skeletal-related event information was collected but will be reported elsewhere. A prostate-specific antigen (PSA) response (defined as the proportion of patients achieving a ≥50% decrease in PSA from baseline to the lowest post-baseline PSA result ($PSA_{50}$), confirmed by a second consecutive PSA assessment at least 3 weeks later. Circulating tumor cell (CTC) conversion rate was defined as the proportion of patients achieving a decline in the number of CTCs from ≥5 cells/7.5 mL at baseline to <5 cells/7.5 mL post-baseline (Veridex™).[6,7] Patients evaluable for PSA response and CTC conversion rate were those with a valid baseline and post-baseline PSA measurement and CTC count measurement, respectively; for CTC conversion analyses, only patients with baseline values ≥5 cells/7.5 mL at baseline were included.

Soft-tissue computed tomography (CT) or magnetic resonance imaging (MRI) of the chest, abdomen, and pelvis, as well as bone scans (using bone scintigraphy commonly performed with technetium-99), were conducted at baseline and then every 8 weeks until disease progression. Criteria for bone and soft-tissue progression are presented in **Supplementary Table S1**. Bone lesions were assessed by bone scans and were separate to the Response Evaluation Criteria in Solid Tumors version 1.1 malignant soft-tissue assessment. In line with the Prostate Cancer Working Group 3 criteria,[8] positive hot spots on the bone scans were considered significant and unequivocal sites of malignant disease and were recorded as metastatic bone lesions.

Blood samples for PSA assessments were collected at baseline, every 4 weeks until week 24, then every 8 weeks until study treatment discontinuation, and for CTC analyses were collected at baseline and then every 8 weeks until study treatment discontinuation. Overall survival was assessed every 12 weeks after progression. Adverse events (AEs) were graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 4.03.

**Supplementary Table S1. Requirements for determination and confirmation of imaging-based progression by either bone scan (bone progression) or CT/MRI (soft-tissue progression)**

| Visit date | Criteria for bone progression | Criteria for soft-tissue progression |
|---|---|---|
| Week 8 | ▪ ≥2 new lesions compared with baseline bone scan<br>▪ Requires confirmation scan ≥6 weeks later with >2 additional lesions compared with week 8 scan | ▪ Progressive disease on CT or MRI by RECIST 1.1<br>▪ No confirmation scan required |
| Week 16 or later | ▪ ≥2 new lesions compared with week 8 bone scan<br>▪ Requires confirmation scan ≥6 weeks later for persistence or increase in number of lesions | ▪ Progressive disease on CT or MRI by RECIST 1.1<br>▪ No confirmation scan required |

CT, computed tomography; MRI, magnetic resonance imaging; RECIST 1.1, Response Evaluation Criteria in Solid Tumors version 1.1

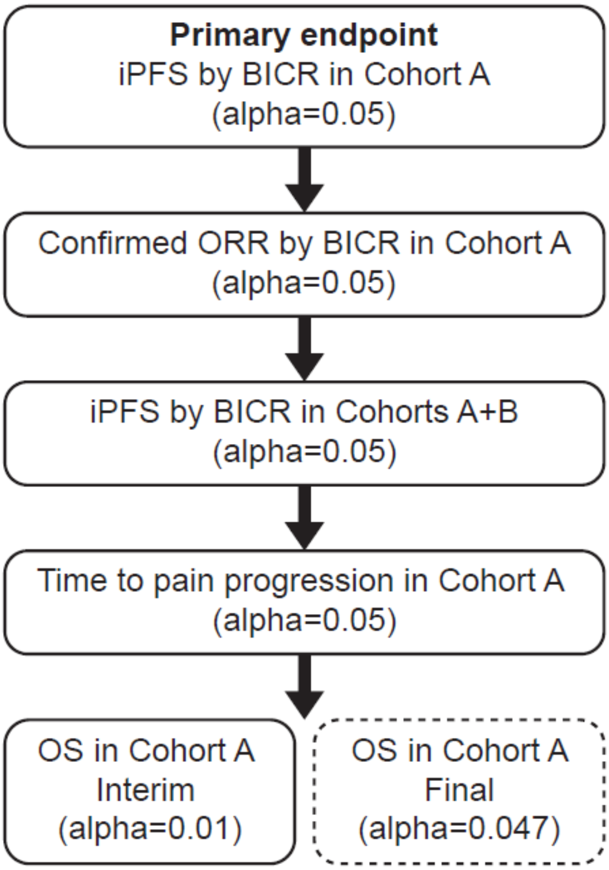
### Subsequent therapies and crossover to olaparib

Subsequent therapies were administered at the investigators' discretion. For patients randomized to the comparator arm who had blinded independent central review (BICR)-confirmed progression, access to olaparib (300 mg twice daily) was allowed if they had not received any subsequent anticancer therapy following discontinuation of randomized treatment and any unresolved toxicities from prior therapy were controlled and were grade ≤1 at the time of initiating olaparib treatment. If patients were not eligible for or chose not to switch to olaparib, subsequent therapies were administered at the investigators' discretion. For patients who were randomized to receive physician's choice of treatment who subsequently switched to olaparib treatment following progression, safety data were summarized according to the treatment they received at the time of the onset of the AE or laboratory result. Tumor assessment by bone scan and CT/MRI was ceased, but follow-up for second progression and survival continued.

### Statistical analyses

The study sponsors were blinded to actual treatment arm until the randomization codes were received on 29 July 2019. Rates of patients achieving a confirmed PSA response or CTC conversion were summarized using descriptive statistics. All reported *P* values are two-sided and coincide with the reported two-sided confidence intervals (CIs). For secondary endpoints where analysis was not adjusted for multiplicity (shown in **Figure S1**), point estimates of treatment effects with 95% CIs are reported (widths of the intervals were not adjusted for multiplicity and the inferences drawn may not be reproducible).

**Supplementary Figure S1. Multiplicity strategy for primary endpoint and key secondary endpoints maintaining overall type I error rate**



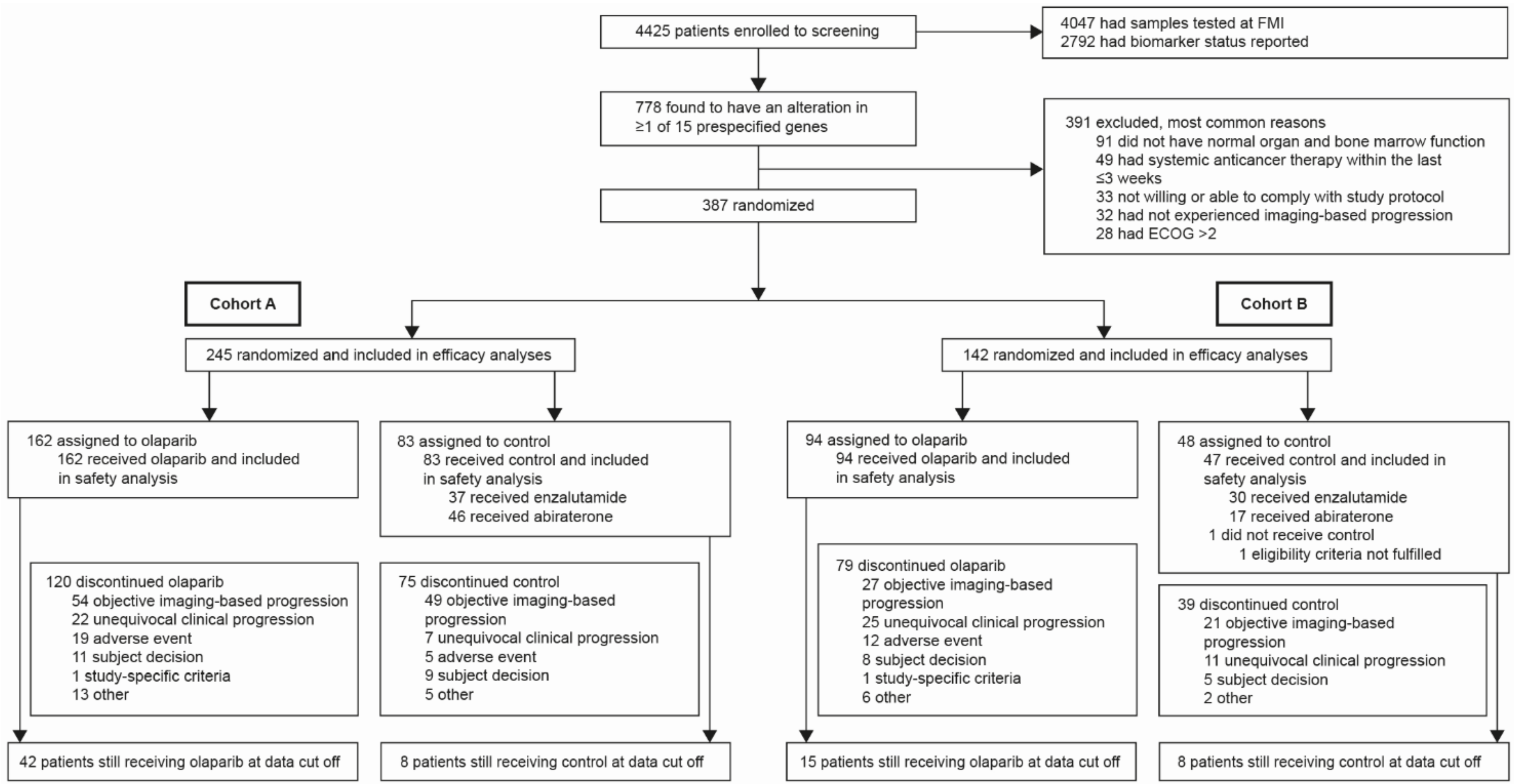

*Alpha was spent at interim and primary OS analyses using an O'Brien-Fleming spending function
BICR, blinded independent central review; ORR, objective response rate; OS, overall survival; iPFS, imaging-based progression-free survival

## Results

### Patient disposition

Two patients with an alteration in *ATM* (both in the olaparib arm), one patient with a *BRCA2* alteration (olaparib arm), and one patient with co-occurring *BRCA2+CDK12* alterations (control arm) were incorrectly assigned to Cohort B; analyses were conducted using data from patients as assigned. One patient randomized to the control arm in Cohort B did not receive treatment.

**Supplementary Figure S2. Enrollment, screening, randomization and intervention in patients**




Testing prior to progression on prior therapy was allowed and enrollment to the study was rapid; this meant that the study was closed before a large proportion of patients with a qualifying alteration detected could be randomized – these patients were considered screen failures at the time of study closing as they did not meet all eligibility criteria, however this related to the testing procedure itself.

ECOG, Eastern Cooperative Oncology Group; FMI, Foundation Medicine

**Prevalence of qualifying alterations in genes with a direct or indirect role in HRR**

**Supplementary Table S2. Prevalence of qualifying gene alterations in randomized patients reported for (A) total number of alterations in any gene and (B) co-occurring alterations**

**(A)**

| | **Cohort A** | | **Cohort B** | | **Overall population (Cohorts A+B)** | |
|---|---|---|---|---|---|---|
| **Patients, n (%)** | **Olaparib 300 mg bid (N=162)** | **Control (N=83)** | **Olaparib 300 mg bid (N=94)** | **Control (N=48)** | **Olaparib 300 mg bid (N=256)** | **Control (N=131)** |
| *BRCA1* | 10 (6.2) | 5 (6.0) | 0 | 0 | 10 (3.9) | 5 (3.8) |
| *BRCA2* | 91 (56.2) | 52 (62.7) | 1 (1.1) | 1 (2.1) | 92 (35.9) | 53 (40.5) |
| *ATM* | 64 (39.5) | 26 (31.3) | 2 (2.1) | 0 | 66 (25.8) | 26 (19.8) |
| *BARD1* | 2 (1.2) | 0 | 1 (1.1) | 1 (2.1) | 3 (1.2) | 1 (0.8) |
| *BRIP1* | 0 | 0 | 2 (2.1) | 2 (4.2) | 2 (0.8) | 2 (1.5) |
| *CDK12* | 3 (1.9) | 2 (2.4) | 64 (68.1) | 30 (62.5) | 67 (26.2) | 32 (24.4) |
| *CHEK1* | 0 | 0 | 2 (2.1) | 1 (2.1) | 2 (0.8) | 1 (0.8) |
| *CHEK2* | 4 (2.5) | 1 (1.2) | 7 (7.4) | 5 (10.4) | 11 (4.3) | 6 (4.6) |
| *FANCL* | 0 | 0 | 0 | 0 | 0 | 0 |
| *PALB2* | 0 | 0 | 4 (4.3) | 4 (8.3) | 4 (1.6) | 4 (3.1) |
| *PPP2R2A* | 1 (0.6) | 3 (3.6) | 6 (6.4) | 5 (10.4) | 7 (2.7) | 8 (6.1) |
| *RAD51B* | 1 (0.6) | 1 (1.2) | 4 (4.3) | 1 (2.1) | 5 (2.0) | 2 (1.5) |
| *RAD51C* | 0 | 0 | 0 | 0 | 0 | 0 |
| *RAD51D* | 1 (0.6) | 0 | 1 (1.1) | 0 | 2 (0.8) | 0 |
| *RAD54L* | 1 (0.6) | 0 | 3 (3.2) | 2 (4.2) | 4 (1.6) | 2 (1.5) |

Patients with multiple genes are included across more than one gene.

**(B)**

| Patients, n (%) | Cohort A | | Cohort B | | Overall population (Cohorts A+B) | |
|---|---|---|---|---|---|---|
| | Olaparib 300 mg bid (N=14) | Control (N=7) | Olaparib 300 mg bid (N=3) | Control (N=4) | Olaparib 300 mg bid (N=17) | Control (N=11) |
| *BRCA1 + ATM* | 1 (7.1) | 0 | 0 | 0 | 1 (5.9) | 0 |
| *BRCA1 + RAD54L* | 1 (7.1) | 0 | 0 | 0 | 1 (5.9) | 0 |
| *BRCA2 + ATM* | 2 (14.3) | 0 | 0 | 0 | 2 (11.8) | 0 |
| *BRCA2 + BARD1* | 2 (14.3) | 0 | 0 | 0 | 2 (11.8) | 0 |
| *BRCA2 + CDK12* | 2 (14.3) | 2 (28.6) | 0 | 1 (25.0) | 2 (11.8) | 3 (27.3) |
| *BRCA2 + CDK12 + CHEK2* | 1 (7.1) | 0 | 0 | 0 | 1 (5.9) | 0 |
| *BRCA2 + CHEK2* | 2 (14.3) | 0 | 0 | 0 | 2 (11.8) | 0 |
| *BRCA2 + CHEK2 + RAD51D* | 1 (7.1) | 0 | 0 | 0 | 1 (5.9) | 0 |
| *BRCA2 + PPP2R2A* | 1 (7.1) | 2 (28.6) | 0 | 0 | 1 (5.9) | 2 (18.2) |
| *BRCA2 + RAD51B* | 0 | 1 (14.3) | 0 | 0 | 0 | 1 (9.1) |
| *ATM + CHEK2* | 0 | 1 (14.3) | 0 | 0 | 0 | 1 (9.1) |
| *ATM + PPP2R2A* | 0 | 1 (14.3) | 0 | 0 | 0 | 1 (9.1) |
| *ATM + RAD51B* | 1 (7.1) | 0 | 0 | 0 | 1 (5.9) | 0 |
| *BARD1 + CDK12* | 0 | 0 | 1 (33.3) | 0 | 1 (5.9) | 0 |
| *BRIP1 + PALB2* | 0 | 0 | 0 | 1 (25.0) | 0 | 1 (9.1) |
| *CDK12 + CHEK1* | 0 | 0 | 1 (33.3) | 0 | 1 (5.9) | 0 |
| *CDK12 + PALB2* | 0 | 0 | 1 (33.3) | 1 (25.0) | 1 (5.9) | 1 (9.1) |
| *PALB2 + PPP2R2A* | 0 | 0 | 0 | 1 (25.0) | 0 | 1 (9.1) |

Only patients with ≥2 qualifying alterations are included. Rows are mutually exclusive.

## Baseline characteristics in patients who were lost to follow-up/withdrew consent

A small number of patients were censored in the imaging-based progression-free survival analysis because they were lost to follow-up/withdrew consent. **Supplementary Table S3** below provides key baseline characteristics for these patients alongside those patients who had follow-up. As the numbers of patients who were lost to follow-up were small, comparisons are challenging to make, and there are no clear differences between these sub-populations.

**Supplementary Table S3. Key baseline characteristics in patients censored in the imaging-based progression-free survival (iPFS) analysis and those with follow-up for (A) Cohort A and (B) the overall population (Cohorts A+B)**

**(A)**

| Cohort A | Patients censored in iPFS analysis due to lost to follow-up | | Patients censored in iPFS analysis for other reasons than lost to follow-up | | Patients in iPFS analysis with follow-up | |
|---|---|---|---|---|---|---|
| | **Olaparib (N=9)** | **Control arm (N=4)** | **Olaparib (N=47)** | **Control arm (N=11)** | **Olaparib (N=153)** | **Control arm (N=79)** |
| Age ≥65 years at randomization, n (%) | 5 (55.6) | 3 (75.0) | 30 (63.8) | 9 (81.8) | 103 (67.3) | 57 (72.2) |
| Baseline PSA, above or equal to medium, n (%) | 4 (44.4) | 3 (75.0) | 14 (29.8) | 5 (45.5) | 64 (41.8) | 45 (57.0) |
| Measurable disease at baseline,† n (%) | 5 (55.6) | 1 (25.0) | 20 (42.6) | 2 (18.2) | 90 (58.8) | 45 (57.0) |
| Metastases at baseline,† n (%) | | | | | | |
| Bone only | 3 (33.3) | 3 (75.0) | 23 (48.9) | 4 (36.4) | 54 (35.3) | 20 (25.3) |
| Visceral (lung/ liver) | 2 (22.2) | 1 (25.0) | 6 (12.8) | 2 (18.2) | 44 (28.8) | 31 (39.2) |
| Other | 3 (33.3) | 0 (0.0) | 11 (23.4) | 4 (36.4) | 46 (30.1) | 23 (29.1) |
| ECOG performance status, n (%) | | | | | | |
| 0 | 3 (33.3) | 2 (50.0) | 25 (53.2) | 9 (81.8) | 81 (52.9) | 32 (40.5) |
| 1 | 6 (66.7) | 2 (50.0) | 20 (42.6) | 2 (18.2) | 61 (39.9) | 44 (55.7) |
| 2 | 0 (0.0) | 0 (0.0) | 2 (4.3) | 0 (0.0) | 11 (7.2) | 3 (3.8) |
| Previous taxane use, n (%) | 6 (66.7) | 1 (25.0) | 28 (59.6) | 4 (36.4) | 100 (65.4) | 51 (64.6) |

**(B)**

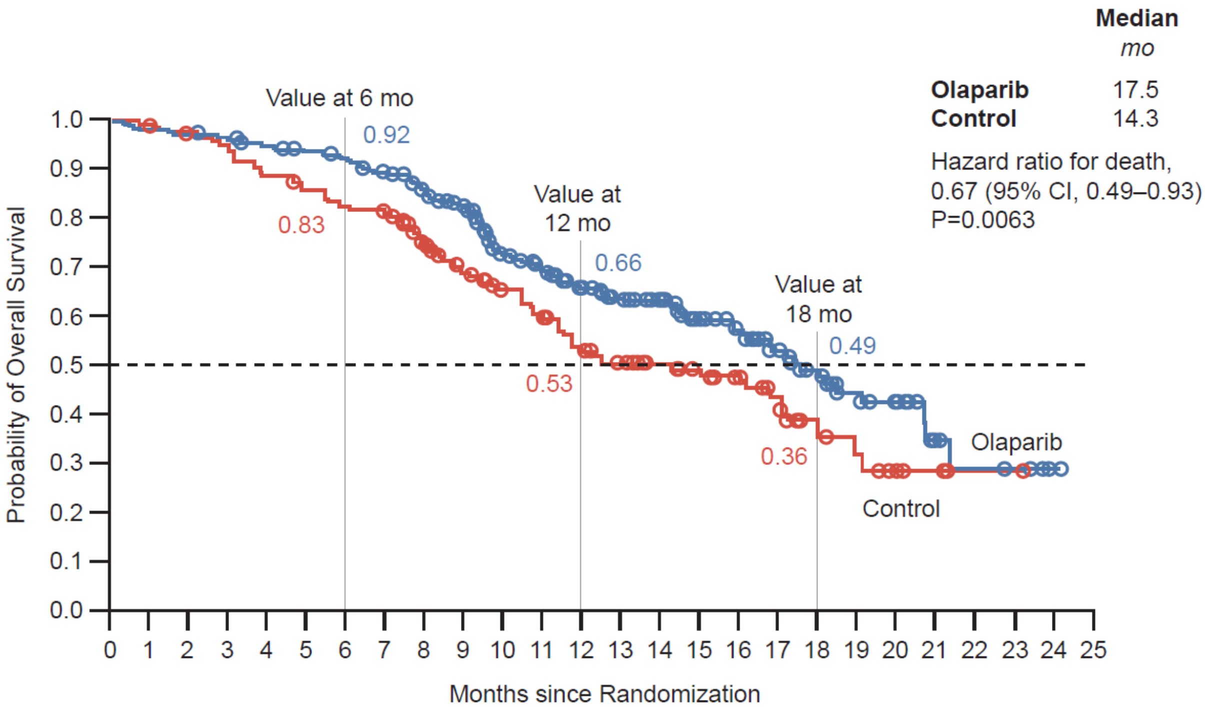
| **Overall population (Cohorts A + B)** | **Patients censored in iPFS analysis due to lost to follow-up** | | **Patients censored in iPFS analysis for other reasons than lost to follow-up** | | **Patients in iPFS analysis with follow-up** | |
|---|---|---|---|---|---|---|
| | **Olaparib (N=10)** | **Control arm (N=8)** | **Olaparib (N=66)** | **Control arm (N=24)** | **Olaparib (N=246)** | **Control arm (N=123)** |
| Age ≥65 years at randomization, n (%) | 6 (60.0) | 7 (87.5) | 44 (66.7) | 19 (79.2) | 168 (68.3) | 90 (73.2) |
| Baseline PSA, above or equal to medium, n (%) | 5 (50.0) | 6 (75.0) | 22 (33.3) | 9 (37.5) | 110 (44.7) | 69 (56.1) |
| Measurable disease at baseline,† n (%) | 5 (50.0) | 4 (50.0) | 26 (39.4) | 5 (20.8) | 144 (58.5) | 68 (55.3) |
| Metastases at baseline,† n (%) | | | | | | |
| Bone only | 4 (40.0) | 3 (37.5) | 31 (47.0) | 11 (45.8) | 82 (33.3) | 35 (28.5) |
| Visceral (lung/ liver) | 2 (20.0) | 2 (25.0) | 7 (10.6) | 3 (12.5) | 66 (26.8) | 42 (34.1) |
| Other | 3 (30.0) | 2 (25.0) | 19 (28.8) | 9 (37.5) | 85 (34.6) | 39 (31.7) |
| ECOG performance status, n (%) | | | | | | |
| 0 | 4 (40.0) | 3 (37.5) | 35 (53.0) | 16 (66.7) | 127 (51.6) | 52 (42.3) |
| 1 | 6 (60.0) | 5 (62.5) | 28 (42.4) | 8 (33.3) | 106 (43.1) | 66 (53.7) |
| 2 | 0 (0.0) | 0 (0.0) | 3 (4.5) | 0 (0.0) | 13 (5.3) | 4 (3.3) |
| Previous taxane use, n (%) | 6 (60.0) | 3 (37.5) | 40 (60.6) | 11 (45.8) | 164 (66.7) | 81 (65.9) |

†Derived from electronic case report forms
ECOG, Eastern Cooperative Oncology Group; PSA, prostate-specific antigen

## Interim overall survival in the overall population (Cohorts A+B)

For censored patients in the overall population (Cohorts A+B), the median duration of follow-up for overall survival was 12.2 months in the olaparib arm and 12.1 months in the control arm.

**Supplementary Figure S3. Kaplan-Meier estimate of interim overall survival in patients with alterations in patients with alterations in *BRCA1*, *BRCA2*, *ATM* and/or any of 12 other prespecified genes (Cohorts A+B; overall population) who received either olaparib or physician's choice of enzalutamide or abiraterone (control)**




No. at Risk

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Olaparib | 256 | 252 | 249 | 245 | 240 | 234 | 227 | 215 | 187 | 167 | 134 | 122 | 106 | 94 | 83 | 67 | 58 | 42 | 34 | 24 | 17 | 7 | 5 | 4 | 1 | 0 |
| Control | 131 | 129 | 125 | 121 | 115 | 110 | 106 | 103 | 79 | 69 | 60 | 55 | 46 | 40 | 34 | 30 | 25 | 20 | 11 | 9 | 6 | 3 | 1 | 1 | 0 | 0 |

CI, confidence interval; mo, months.

## Sensitivity analysis for ascertainment bias

**Supplementary Table S4. Prespecified sensitivity analyses of imaging-based progression-free survival (iPFS) by investigator assessment**

| | Patients with alterations in *BRCA1, BRCA2* and/or *ATM* (Cohort A) | | Patients with an alteration in one of the other 12 genes (Cohort B) | | Overall population (Cohorts A+B) | |
|---|---|---|---|---|---|---|
| | **Olaparib (N=162)** | **Control arm (N=83)** | **Olaparib (N=94)** | **Control arm (N=48)** | **Olaparib (N= 256)** | **Control arm (N=131)** |
| Progression or death, n (%) | 95 (58.6) | 66 (79.5) | 68 (72.3) | 33 (68.8) | 163 (63.7) | 99 (75.6) |
| Median iPFS (months) | 9.8 | 3.6 | 5.6 | 3.4 | 7.5 | 3.5 |
| Hazard ratio (95% CI) | 0.24 (0.17, 0.34) | | 0.60 (0.39, 0.93) | | 0.36 (0.27, 0.47) | |

CI, confidence interval

### Sensitivity analysis for missed visits

The statistical analysis plan for the primary analysis prespecified that if a patient's disease progressed or they died immediately after two or more consecutive missed visits for either soft tissue or bone assessments, the patient would be censored at the earliest of time of either the previous Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST 1.1) assessment (taking the latest target lesion, non-target lesion or new lesion scan date) or previous bone scan assessment prior to the two consecutive missed visits (if RECIST 1.1 assessment and bone scan were done at different visits). An exploratory sensitivity analysis was conducted to consider the impact of censoring these patients on imaging-based progression-free survival, which included all progression or death after two or more missed visits as events. The results, presented in **Supplementary Table S5**, are consistent with the prespecified primary analysis.

**Supplementary Table S5. Sensitivity analysis for imaging-based progression-free survival (iPFS) by blinded independent central review without the "two missed visits" rule**

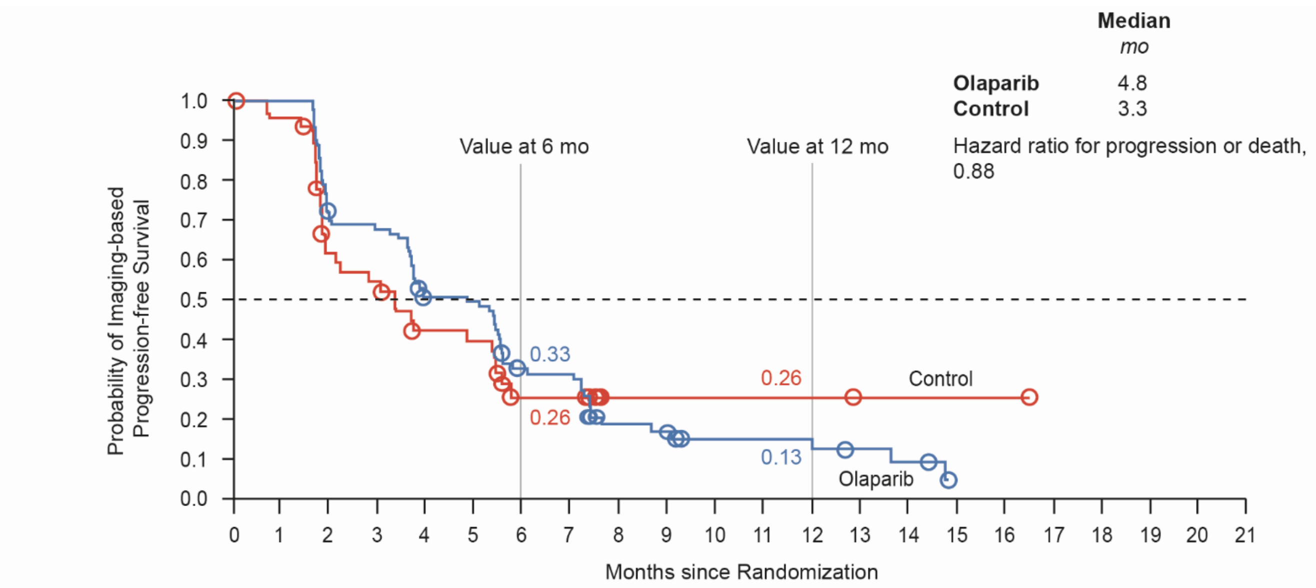
| **Cohort A** | **Olaparib (n=162)** | **Control arm (n=83)** |
|---|---|---|
| Events, n (%) | 113 (69.8) | 70 (84.3) |
| Median iPFS (months) | 7.6 | 3.6 |
| Hazard ratio (95% CI) | 0.361 (0.264, 0.496) | |
| P value | <0.0001 | |
| **Overall population (Cohorts A + B)** | **Olaparib (n=256)** | **Control arm (n=131)** |
| Events, n (%) | 189 (73.8) | 104 (79.4) |
| Median iPFS (months) | 6.2 | 3.5 |
| Hazard ratio (95% CI) | 0.506 (0.397, 0.649) | |
| P value | <0.0001 | |

CI, confidence interval

## Exploratory analyses for genes other than *BRCA1*, *BRCA2* and/or *ATM* (Cohort B)

**Supplementary Figure S4. (A) Imaging-based progression-free survival by blinded independent central review and (B) overall survival among patients with an alteration in genes other than *BRCA1*, *BRCA2*, or *ATM* (Cohort B)**

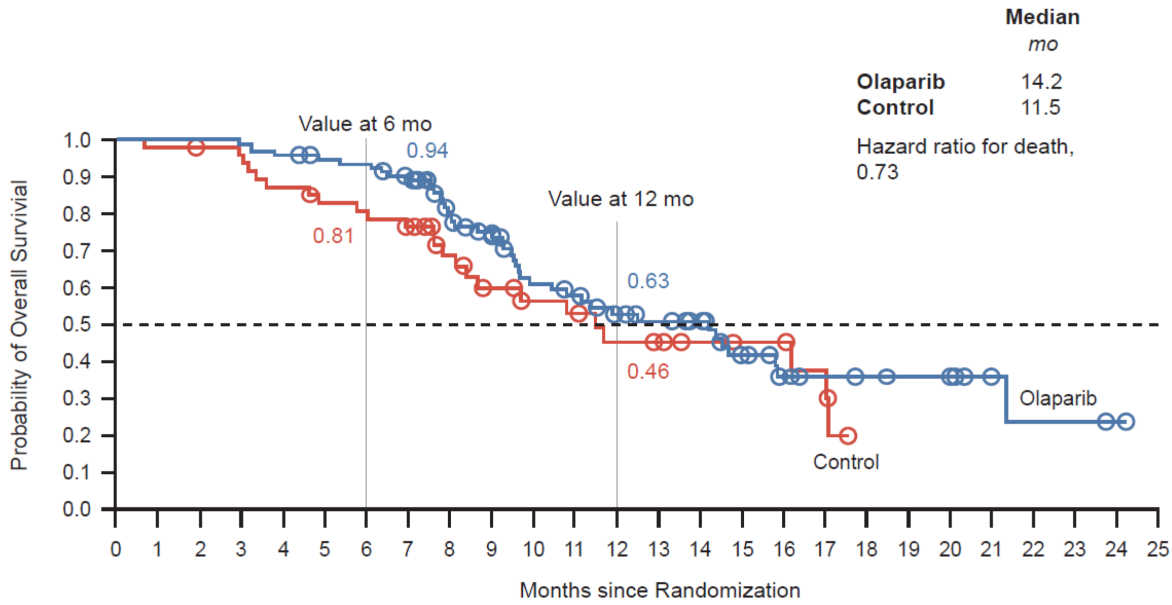
**(A)**




No. at Risk

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Olaparib | 94 | 90 | 62 | 60 | 43 | 42 | 24 | 23 | 11 | 10 | 6 | 6 | 5 | 4 | 3 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Control | 48 | 44 | 26 | 23 | 16 | 15 | 7 | 7 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |

**(B)**




No. at Risk

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Olaparib | 94 | 94 | 94 | 93 | 90 | 87 | 86 | 79 | 62 | 52 | 39 | 36 | 30 | 27 | 24 | 17 | 12 | 9 | 8 | 7 | 6 | 3 | 2 | 2 | 1 | 0 |
| Control | 48 | 47 | 46 | 45 | 41 | 38 | 37 | 34 | 25 | 19 | 16 | 15 | 12 | 11 | 9 | 7 | 7 | 5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

CI, confidence interval; mo, months

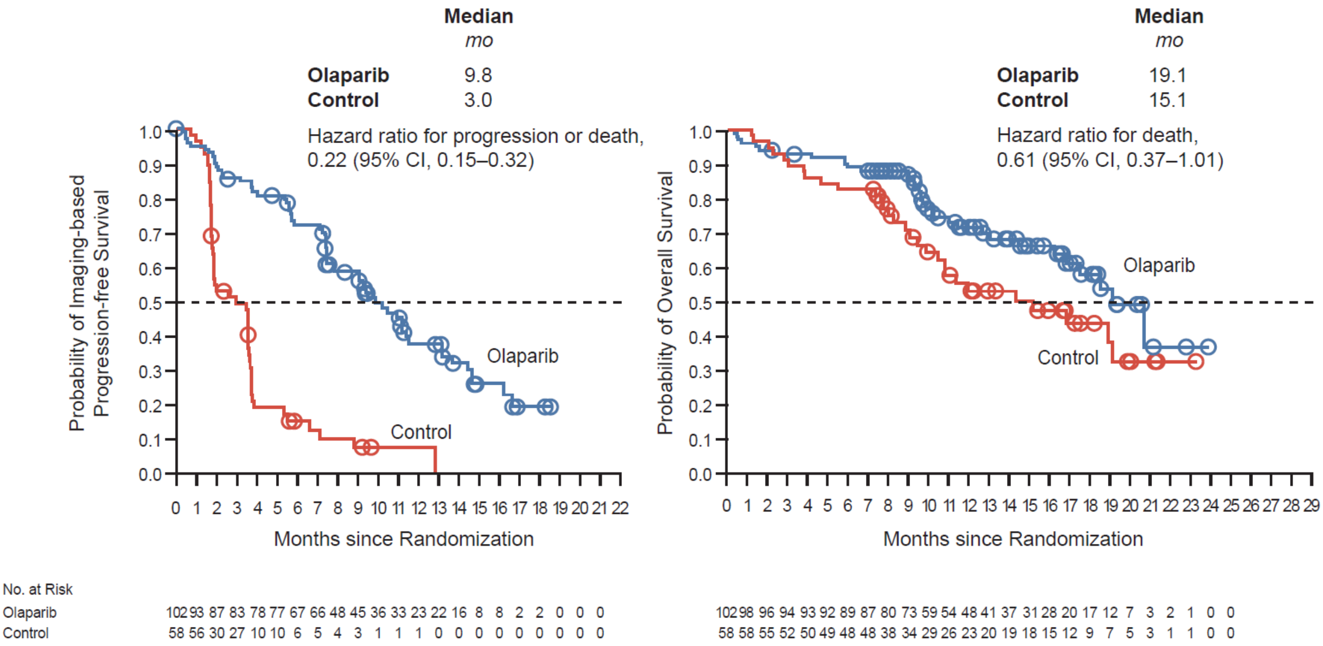
## Exploratory analyses at the gene level

Gene-level analysis is complex and exploratory – comparisons may be confounded by small sample size, low power, lack of stratification at the gene level and limited knowledge of the performance on prior, and standard of care therapies. In this study, 97% of patients were randomized based on alterations in eight of the 15 genes; 7/15 genes therefore had alteration frequencies too low for descriptive statistics. In exploratory analyses at the gene level there was evidence of clinical activity of olaparib in patients with alterations in genes other than *BRCA1* or *BRCA2*, and further investigation is warranted (**Supplementary Figure S5**).

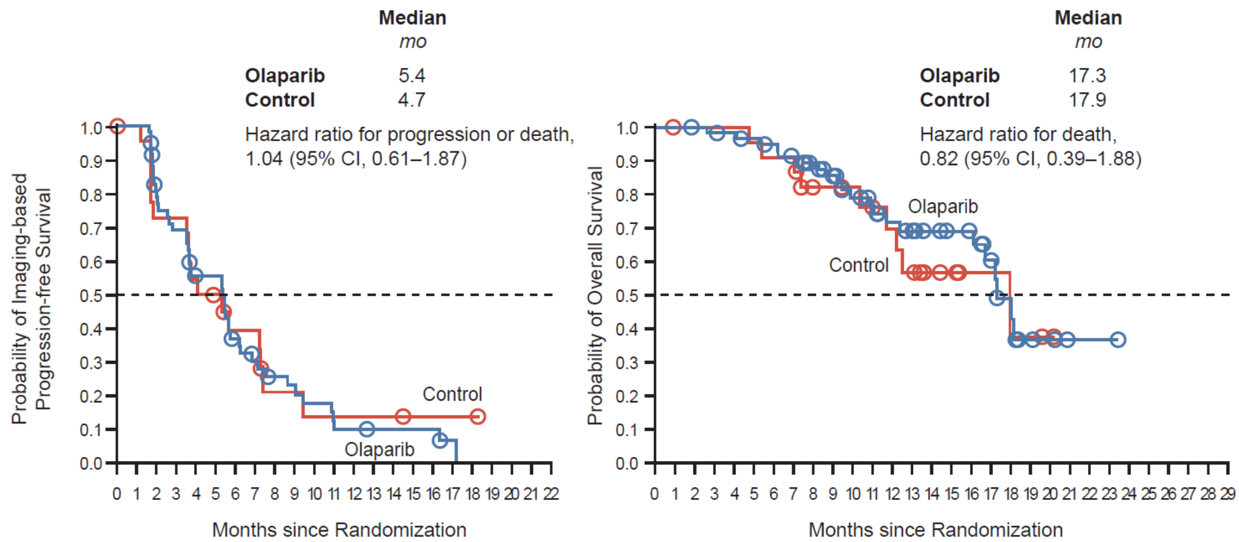
Exploratory analyses showed that the median imaging-based progression-free survival for the *ATM* subgroup was 5.36 in the olaparib arm and 4.7 months in the control arm (**Supplementary Figure S4**). For the control arm, this was longer than observed in the *BRCA1* or *BRCA2* (3.0 months) and *CDK12* (2.2 months) subgroups. Additional exploratory analysis revealed no difference in baseline characteristics in the *ATM* subgroup compared to the other gene subgroups. However, when examining *ATM* patients by taxane history, potential differences were revealed, as the control arm in the taxane-naïve group appeared to behave differently (**Supplementary Table S6**). These observations in a very small number of patients can be considered hypothesis-generating.

**Supplementary Figure S5. Imaging-based progression-free survival by blinded independent central review and overall survival among patients in the overall population with an alteration* in (A) *BRCA1* and/or *BRCA2*, (B) *ATM* or (C) *CDK12***

### (A) *BRCA1* and/or *BRCA2*



## (B)

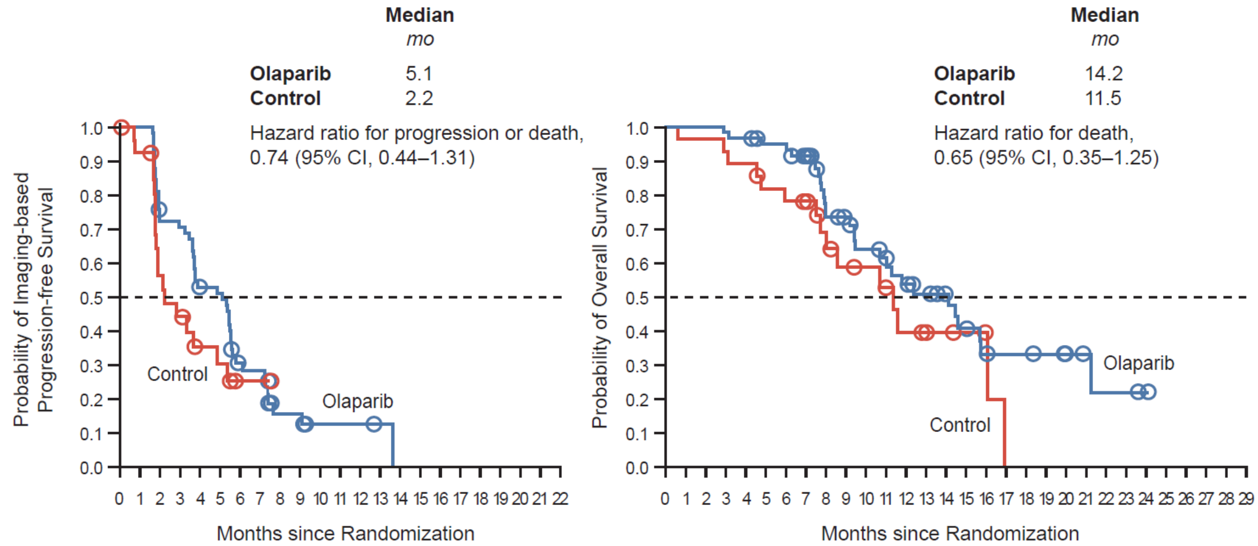



No. at Risk

| | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Olaparib | 62 | 59 | 42 | 36 | 27 | 27 | 17 | 13 | 10 | 9 | 7 | 5 | 4 | 3 | 3 | 3 | 3 | 1 | 0 | 0 | 0 | 0 |
| Control | 24 | 22 | 16 | 16 | 12 | 10 | 7 | 7 | 3 | 3 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |

| | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Olaparib | 62 | 62 | 61 | 60 | 59 | 57 | 54 | 51 | 47 | 43 | 36 | 32 | 28 | 26 | 22 | 19 | 18 | 12 | 8 | 4 | 3 | 1 | 1 | 1 | 0 | 0 |
| Control | 24 | 23 | 23 | 23 | 23 | 22 | 21 | 21 | 15 | 15 | 14 | 13 | 11 | 9 | 6 | 5 | 3 | 3 | 2 | 2 | 1 | 0 | 0 | 0 | 0 | 0 |

## (C)




No. at Risk

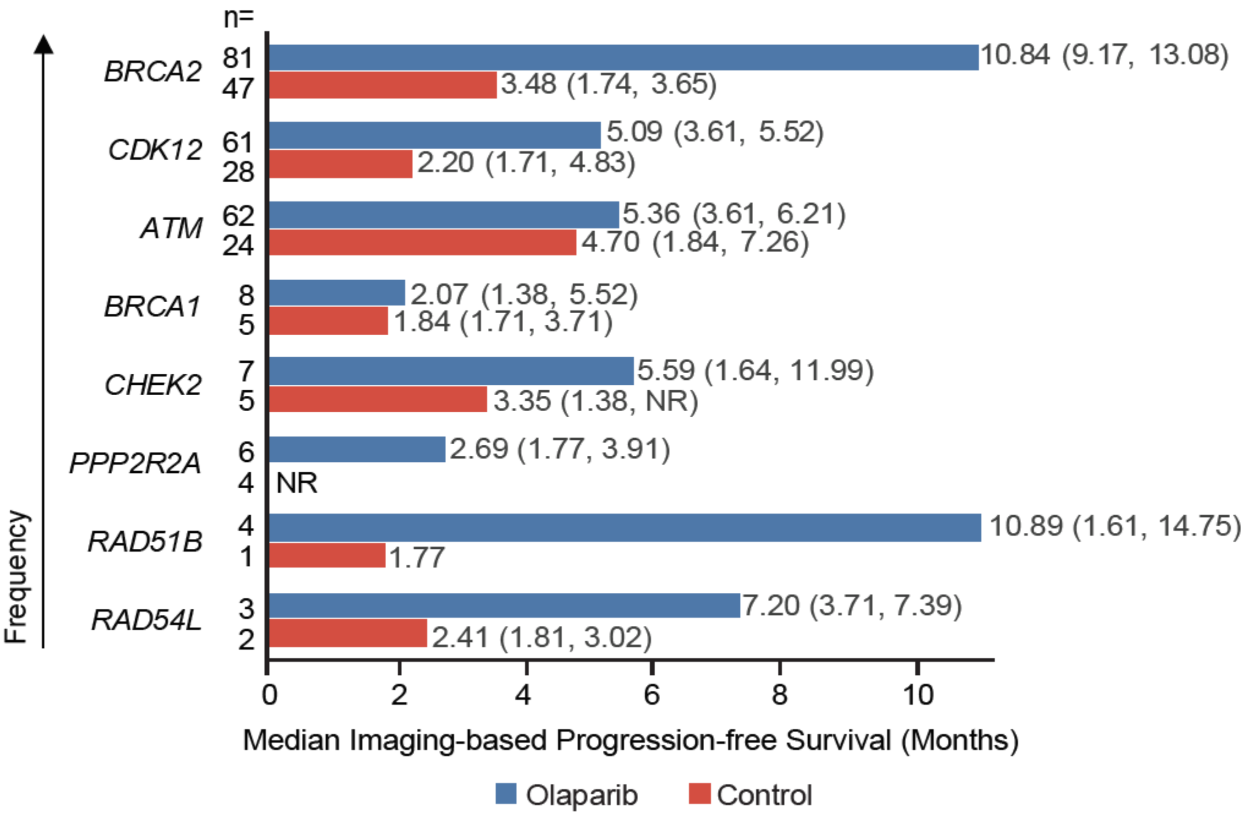
| | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Olaparib | 61 | 58 | 41 | 40 | 29 | 28 | 13 | 12 | 5 | 5 | 2 | 2 | 2 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Control | 28 | 24 | 14 | 11 | 7 | 6 | 3 | 3 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Olaparib | 61 | 61 | 61 | 60 | 59 | 56 | 56 | 52 | 39 | 33 | 27 | 25 | 21 | 18 | 16 | 12 | 9 | 7 | 7 | 6 | 5 | 3 | 2 | 2 | 1 | 0 |
| Control | 28 | 27 | 27 | 26 | 25 | 22 | 22 | 20 | 14 | 11 | 10 | 9 | 6 | 5 | 4 | 3 | 3 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

*Includes patients with an alteration in a single gene

CI, confidence interval; mo, months

**Supplementary Figure S6. Median (95% confidence interval) imaging-based progression-free survival by blinded independent central review by gene subgroups in patients with alterations in a single gene***




*Only includes gene subgroups with at least five patients randomized

NR, not reached

**Supplementary Table S6. Imaging-based progression-free survival (iPFS) by blinded independent central review (BICR) and overall survival (OS) among patients with an alteration in *ATM* by taxane treatment history***

| | Prior taxane | | No prior taxane | |
|---|---|---|---|---|
| | **Olaparib (n=35)** | **Control arm (n=15)** | **Olaparib (n=27)** | **Control arm (n=9)** |
| **iPFS by BICR** | | | | |
| Events, n (%) | 25 (71.4) | 12 (80.0) | 21 (77.8) | 5 (55.6) |
| Median (months) | 5.7 | 3.6 | 5.3 | 5.7 |
| Hazard ratio (95% CI) | 0.552 (0.280, 1.149) | | 2.408 (0.969, 7.295) | |
| **OS** | | | | |
| Events, n (%) | 12 (34.3) | 8 (53.3) | 9 (33.3) | 1 (11.1) |
| Median (months) | 17.2 | 12.5 | NR | NR |
| Hazard ratio (95% CI) | 0.544 (0.223, 1.399) | | 2.943 (0.552, 54.292) | |

NR, not reached

*Includes patients with an alteration in *ATM* only

### Sensitivity analysis for time to pain progression

An exploratory sensitivity analysis was conducted to evaluate the impact of competing risk of death on the analysis of time to pain progression. Specifically, any death in the absence of pain progression was counted as an event. The results, presented in **Supplementary Table S7**, are consistent with those of the prespecified analysis.

**Supplementary Table S7. Time to pain progression (TTPP) sensitivity analysis, including any death as an event in the absence of pain progression**

| **Cohort A** | **Olaparib (n=162)** | **Control arm (n=83)** |
|---|---|---|
| Median TTPP (months) | 17.2 | 5.5 |
| Hazard ratio (95% CI) | 0.501 (0.303, 0.849) | |
| P value | 0.0068 | |
| **Overall population (Cohorts A + B)** | **Olaparib (n=256)** | **Control arm (n=131)** |
| Median TTPP (months) | 11.9 | 6.9 |
| Hazard ratio (95% CI) | 0.596 (0.395, 0.917) | |
| P value | 0.0124 | |

CI, confidence interval

### Patients who crossed over to olaparib from the control arm

Overall, 72 patients assigned to the control arm whose disease progressed by BICR crossed over to olaparib (Cohort A, 50/62 [80.6%]; Cohort B, 22/26 [84.6%], Total, 72/88 [81.8%]). The median duration of olaparib treatment in crossover patients was 3.5 months. An additional three patients assigned to the control arm went on to receive olaparib outside the study and were included in sensitivity analyses for the effect of switching to olaparib on overall survival (**Supplementary Table S8**). The method for these analyses used rank preserving structural failure time models and only adjusted for patients on the control arm who subsequently received olaparib, with a re-censoring approach selected as main overall survival treatment switch method. Method (a) in **Supplementary Table S8** follows NICE guidelines for switching adjusted analyses with an active control arm, where a continuing treatment effect is assumed; i.e. patients on the olaparib arm are always considered to be "on" treatment and patients that switch remain "on" treatment from the time of switch until death.[9] A more conservative approach has been applied in method (b). Patients on the olaparib arm are always considered to be "on" treatment, however, patients who switch are assumed to only receive benefit for the time they were receiving olaparib. This leads to a smaller adjustment for the crossover effect and is a method where the assumptions on the control arm differ to those on the active arm. Results show a directional improvement in favor of olaparib in both Cohorts and the overall population.

**Supplementary Table S8. Sensitivity analyses for crossover effect on overall survival results**

| | | Cohort A | Cohort B | Overall population (Cohorts A + B) |
|---|---|---|---|---|
| **Patients switching to olaparib from control arm, n (%)** | | 51 (61.4) | 24 (50.0) | 75 (57.3) |
| **Hazard ratio (95% CI)** | | | | |
| Full analysis set | | 0.64 (0.43, 0.97) | 0.73 (0.45, 1.23) | 0.67 (0.49, 0.93) |
| Treatment switch adjusted | (a) | 0.45 (0.22, 0.94) | 0.40 (0.09, 1.79) | 0.45 (0.24, 0.85) |
| | (b) | 0.57 (0.34, 0.96) | 0.57 (0.23, 1.42) | 0.60 (0.40, 0.90) |

CI, confidence interval

## Safety data

AEs leading to death in the olaparib arm were lung infection (n=2), cardiopulmonary failure (n=2), pneumonia, acute cardiac failure, Budd Chiari syndrome, intestinal diverticulum, neutropenia, septic shock, sudden death, spinal stenosis and urinary retention (all n=1). In the control arm, pneumonia, cardiac failure, arterial thrombosis, pleural effusion and death (all n=1) were reported as AEs leading to death. For the death reported to be related to olaparib treatment, a 56-year-old patient in Cohort A with a history of multiple bone metastases and radiotherapy developed grade 5 lung infection, neutropenia and grade 4 thrombocytopenia 3 days after starting study treatment (he received concomitant dexamethasone from Day 0 to Day 2 for lumbar pain). The patient died 5 days later despite receiving combined antibiotic and antifungal treatment in ICU. For the death reported related to study treatment in the control arm, a 67-year-old male in Cohort A received abiraterone and prednisolone. He had pleural effusion before being randomized to the study and developed worsening of pleural effusion on Day 194 with symptoms of grade 3 dyspnea. Abiraterone and prednisolone were discontinued. Pleural drainage was performed, however 9 days later his dyspnea worsened, and CT described a right main bronchial occlusion. The patient developed respiratory insufficiency requiring intubation.

New primary malignancies were reported as follows; in the olaparib arm, one patient developed glioma (which did not resolve); in the control arm, one patient developed transitional cell carcinoma on treatment Day 227, and one patient developed gastric cancer on treatment Day 85 (both patients were assigned enzalutamide, and both events had resolved by data cut-off). Pneumonitis was reported in four olaparib patients (one Grade 3, two Grade 2, one Grade 1) of which two had resolved at data cut-off for this analysis. One control arm (abiraterone) patient experienced pneumonitis of Grade 2 severity, and one patient (enzalutamide) experienced Grade 1 radiation pneumonitis, one of which had resolved at the time of data cut-off. There was one incidence of interstitial lung disease (Grade 3) in an olaparib patient which then resolved.

*Pulmonary embolism*

Of the 11 (4.3%) pulmonary embolism events reported in the olaparib arm, five were Grade 1 or 2, five were Grade 3, and one was Grade 4; none were fatal. Eight of these 11 patients continued olaparib at unchanged dose and had no repeated pulmonary embolism events. The event in the control arm was Grade 3. Pulmonary embolism is not a recognized complication of olaparib treatment. With over half of these events denoted Grade 1 to 2, and continuation of olaparib therapy in most cases, the significance of the occurrence of these events is difficult to interpret in this highly confounded patient population. There is no plausible biological explanation and no pattern of time to event onset, therefore no causal association can be established.

**Supplementary Table S9. Grade ≥3 adverse events occurring in >2% of patients in the overall population (Cohorts A+B)**

| | Olaparib (N= 256) | Control arm (N=130) |
|---|---|---|
| Any | 130 (50.8) | 49 (37.7) |
| Anemia | 55 (21.5) | 7 (5.4) |
| Neutropenia | 10 (3.9) | 0 |
| Thrombocytopenia | 9 (3.5) | 0 |
| Fatigue/ asthenia | 7 (2.7) | 7 (5.4) |
| Pneumonia | 6 (2.3) | 1 (0.8) |
| Dyspnea | 6 (2.3) | 0 |
| Vomiting | 6 (2.3) | 1 (0.8) |
| Pulmonary embolism | 6 (2.3) | 1 (0.8) |
| Urinary tract infection | 4 (1.6) | 5 (3.8) |
| Hypertension | 3 (1.2) | 3 (2.3) |
| Sepsis | 3 (1.2) | 3 (2.3) |

**Supplementary Table S10. Treatments received by patients in the overall population (Cohorts A+B) following discontinuation of study treatment**

| Subsequent therapy – no. (%) | Olaparib 300 mg bid (N=256) | Control arm (N=130) |
|---|---|---|
| Continuing study treatment | 57 (22.3) | 16 (12.3) |
| Any subsequent therapy | 90 (35.2) | 83 (63.4) |
| Anti-cancer therapies received by ≥2% patients in either arm | | |
| Olaparib | 3 (1.2) | 75 (57.3) |
| Docetaxel | 29 (11.3) | 8 (6.1) |
| Cabazitaxel | 21 (8.2) | 9 (6.9) |
| Abiraterone | 16 (6.3) | 2 (1.5) |
| Enzalutamide | 19 (7.4) | 1 (0.8) |
| Carboplatin | 11 (4.3) | 3 (2.3) |
| Pembrolizumab | 5 (2) | 0 |
| Lutetium 177 | 5 (2) | 0 |

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